Chemistry

Time: 3 Hrs **Total Marks: 70**

Note:

i. All questions are compulsory.
ii. Answer to both sections should be written in the same answer book.
iii. Figure to the right hand side indicate full marks.
iv. Draw neat, labelled diagrams and write balanced equations wherever necessary.
v. Use of logarithmic table is allowed.
vi. Answer to every new question must be started on a new page.

SECTION – I

Q.1. Select and write the most appropriate answer from the given alternatives for each sub-question: **[7]**

i. In Van Arkel method of refining metal, impure zirconium is converted to unstable volatile compound by heating it with _____.
(A) oxygen (B) chlorine
(C) bromine (D) iodine

ii. The temperature at which vapour pressure of a liquid becomes equal to the atmospheric pressure is _____.
(A) melting point (B) boiling point
(C) 273 K (D) 373 K

iii. Which of the following parameters are correct for triclinic lattice?
(A) $\alpha = \beta = \gamma = 90°$ and $a = b = c$ (B) $\alpha \neq \beta \neq \gamma = 90°$ and $a \neq b \neq c$
(C) $\alpha = \gamma = 90°$, $\beta \neq 90°$ and $a \neq b \neq c$ (D) $\alpha \neq \beta \neq \gamma \neq 90°$ and $a \neq b \neq c$

iv. Which mixture is used for respiration by deep sea divers?
(A) $He + O_2$ (B) $Ne + O_2$
(C) $Ar + O_2$ (D) $Kr + O_2$

v. For the reaction, $2N_2O_{5(g)} \longrightarrow 4NO_{2(g)} + O_{2(g)}$ in liquid bromine, which of the following rate equation is INCORRECT?
(A) $-\frac{1}{2}\frac{d[N_2O_5]}{dt}$ (B) $-\frac{1}{4}\frac{d[NO_2]}{dt}$
(C) $\frac{d[O_2]}{dt}$ (D) $\frac{1}{4}\frac{d[NO_2]}{dt}$

vi. For a certain reaction, $\Delta H = -50$ kJ and $\Delta S = -80$ J K^{-1}, at what temperature does the reaction turn from spontaneous to non-spontaneous?
(A) 6.25 K (B) 62.5 K
(C) 625 K (D) 6250 K

vii. What is the ratio of volumes of H_2 and O_2 liberated during electrolysis of acidified water?
(A) 1 : 2 (B) 2 : 1
(C) 1 : 8 (D) 8 : 1

Q.2. Answer any THREE of the following: **[9]**

i. Calculate ΔH° for the following reaction:

$2H_3BO_{3(aq)} \longrightarrow B_2O_{3(s)} + 3H_2O_{(l)}$

Given that,

a. $H_3BO_{3(aq)} \longrightarrow HBO_{2(aq)} + H_2O_{(l)}$, $\Delta H_1^{\circ} = -0.02$ kJ

b. $H_2B_4O_{7(s)} \longrightarrow 2B_2O_{3(s)} + H_2O_{(l)}$, $\Delta H_2^{\circ} = 17.3$ kJ

c. $H_2B_4O_{7(s)} + H_2O_{(l)} \longrightarrow 4HBO_{2(aq)}$, $\Delta H_3^{\circ} = -11.58$ kJ

ii. Calculate molarity and molality of 6.3% solution of nitric acid having density 1.04 g cm^{-3}. (H = 1, N = 14, O = 16)

iii. What is the action of chlorine (Cl) on the following:

a. Cold and dilute caustic soda

b. Hot and concentrated caustic soda

c. Potassium bromide solution

iv. Calculate the number of atoms present in 2 gram of crystal which has face-centred cubic (FCC) crystal lattice having edge length of 100 pm and density 10 gcm^{-3}.

Q.3. Answer any SIX of the following: **[12]**

i. NF_3 is possible, but NF_5 is not. Why?

ii. State and explain Faraday's second law of electrolysis.

iii. Describe 'froth floatation process' for concentration of sulphide ore.

iv. Distinguish between molecularity and order of reaction.

v. What are the conditions for spontaneous and non-spontaneous reactions in terms of free energy change? Define entropy.

vi. Draw the structure of $H_4P_2O_6$ hypophosphoric acid. What is the action of heat on potassium permanganate?

vii. State Henry's law. How does solubility of a gas in water varies with the temperature?

viii. What are Schottky defect and Frenkel defect?

Q.4. Answer any ONE of the following: **[7]**

i. a. Write electrode reaction and net cell reaction for fuel cell. Calculate e.m.f. of the following cell at 25 °C.

$$Zn_{(s)} \,|\, \underset{0.1M}{Zn^{++}_{(aq)}} \,\|\, \underset{0.5M}{Cu^{++}_{(aq)}} \,|\, Cu_{(s)}$$

Standard reduction potential (SRP) of Zn and Cu are – 0.76 V and 0.334 V respectively.

b. Define isotonic solutions.

c. Derive the relation $\Delta H - \Delta U = \Delta nRT$.

ii. a. Define activation energy. Calculate activation energy for a reaction of which rate constant becomes four times when temperature changes from 30 °C to 50 °C. (Given R = 8.314 JK^{-1} mol^{-1}).

b. Draw a neat, well labelled diagram of electrolytic cell for extraction of aluminium.

c. Write electronic configuration and two uses of neon. (Z = 10).

SECTION – II

Q.5. Select and write the most appropriate answer: **[7]**

i. When KOH solution is added to potassium dichromate solution the colour of solution changes to yellow, because _____.

(A) chromate ion changes to dichromate ion
(B) dichromate ion changes to chromate ion
(C) oxidation number of chromium changes from + 6 to + 4
(D) oxidation number of chromium changes from + 4 to +6

ii. But-1-ene on reaction with HCl in the presence of sodium peroxide yields _____.

(A) n-butyl chloride
(B) isobutyl chloride
(C) secondary butyl chloride
(D) tertiary butyl chloride

iii. 3-Methylbutane-2-ol on heating with HI gives _____.

(A) 2-iodo-3-methylbutane (B) 2-iodo-2-methylbutane
(C) 1-iodo-3-methylbutane (D) 1-iodo-2-methylbutane

iv. IUPAC name of $C_6H_5 - CH_2 - \overset{\overset{O}{\|}}{C} - CH_2 - CH_2 - CH_2 - CH_3$ is _____.

(A) 1-Phenylhexan-2-one (B) 6-Phenylhexan-5-one
(C) 1-Benzylhexan-5-one (D) Dodecan-5-one

v. Iodoform is used as an _____.

(A) antiseptic (B) antibiotic
(C) insecticide (D) anaesthetic

vi. Stachyose is an example of _____.

(A) monosaccharides (B) disaccharides
(C) trisaccharides (D) tetrasaccharides

vii. The Zieglar-Natta catalyst is used in the preparation of _____.

(A) LDPE (B) PHBV
(C) PAN (D) HDPE

Q.6. Answer any THREE of the following: **[9]**

i. How is phenol converted into the following?

a. benzene
b. benzoquinone
c. picric acid

ii. Explain the mechanism of aldol addition reaction.

iii. Enlist the properties of glucose that can not be explained on the basis of open chain structure of it.

iv. How is nitromethane prepared from the following?

a. alkyl halide
b. α-halogen carboxylic acid
c. α-nitroalkene

Q.7. Answer any SIX of the following: **[12]**

i. How is methoxy benzene prepared from carbolic acid?

ii. State the superiority of crystal field theory over valence bond theory.

iii. How is benzophenone prepared from benzonitrile?

iv. Explain Hoffmann bromamide degradation reaction.

v. What are hormones? State the function of insulin.

vi. How are polymers classified on the basis of polymerisation process?

vii. Describe 'any two' chemical methods of food preservation.

viii. Write observed electronic configuration of elements from first transition series having half filled d-orbitals.

Q.8. Answer any ONE of the following: **[7]**

i. a. What is lanthanoid contraction? Explain the cause and effects of lanthanoid contraction.

b. Write the structure of melamine.

c. Explain the mechanism of cleaning action of soap.

ii. a. Explain optical activity of lactic acid.

b. Draw a neat, labelled energy profile diagram for SN^1 reaction mechanism.

c. Write applications of co-ordination compounds in medicine and electroplating.

d. Explain the structure of carbonyl functional group.